3.4.4 Capacitance

- ### Capacitance

  Definition of capacitance;

  $$C = \frac{Q}{V}$$

- ### Energy stored by a capacitor

  Derivation of $E = \frac{1}{2}\ Q\ V$ and interpretation of area under a graph of charge against pd

  $E = \frac{1}{2}\ Q\ V = \frac{1}{2}\ C\,V^2 = \frac{1}{2}\ Q^2/\,C$

- ### Capacitor discharge

  Graphical representation of charging and discharging of capacitors through resistors

  Time constant = RC

  Calculation of time constants including their determination from graphical data.

  Quantitative treatment of capacitor discharge

  $Q = Q_o\,e^{-\,t/RC}$

  Candidates should have experience of the use of a voltage sensor and datalogger to plot discharge curves for a capacitor.